Multiple Access

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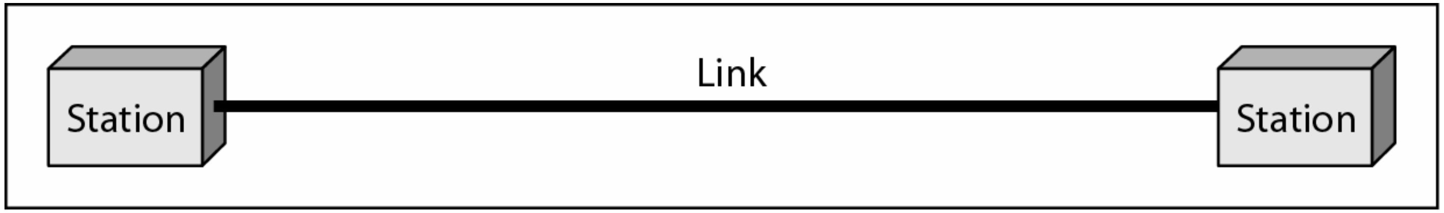
Book chapters

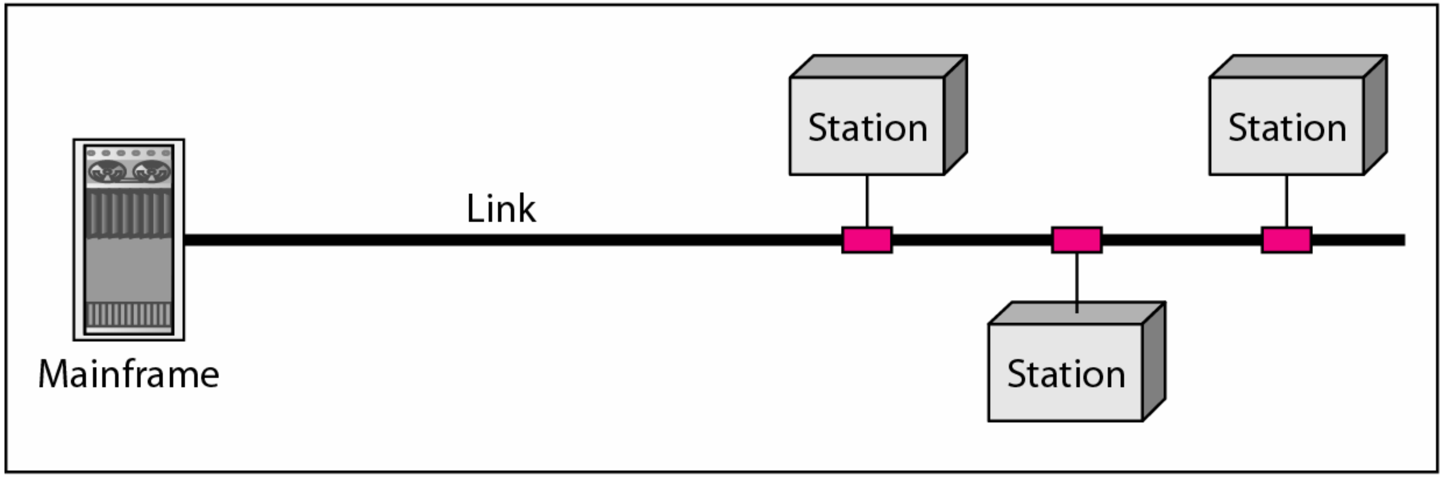
Forouzan: 1.2, 12

Kihl: 5.1, 5.2

Types of connections



a. Point-to-point



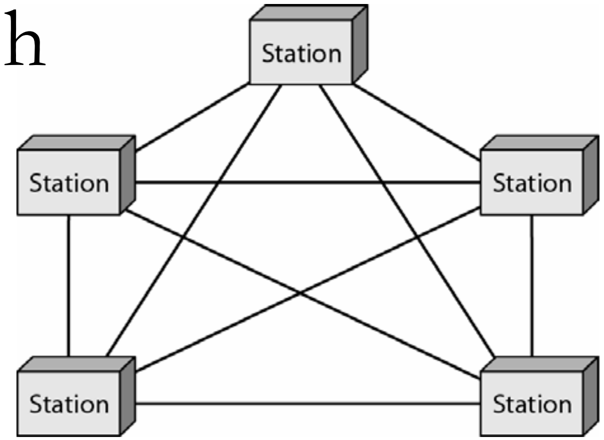
b. Multipoint

Local Area Networks (LAN)

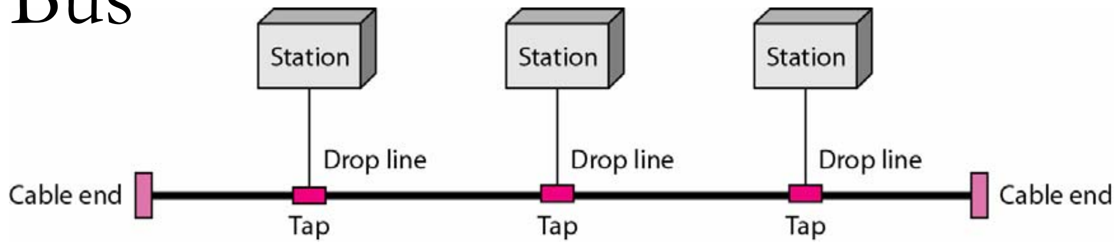
- A Local Area Network (LAN) is usually privately owned and links the devices in a single office, building or campus.
- LAN size is limited to a few kilometers.
- Traditionally, LANs use a shared medium, which means that the stations share a common physical link.
- Wireless LANs (WLANs) are common everywhere.

Wired LAN topologies

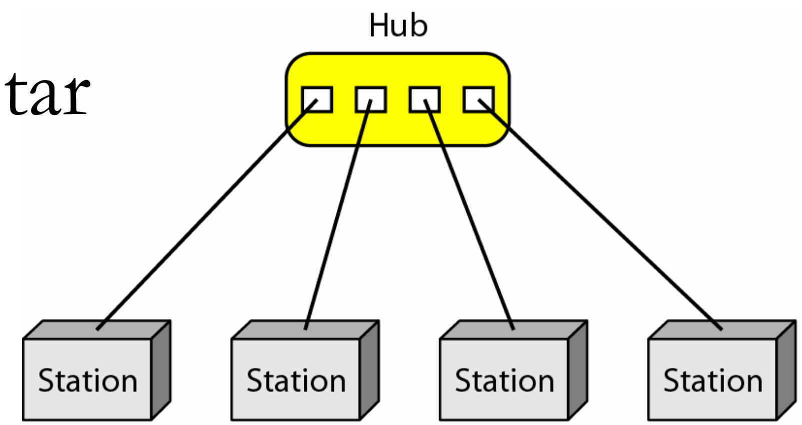
Mesh

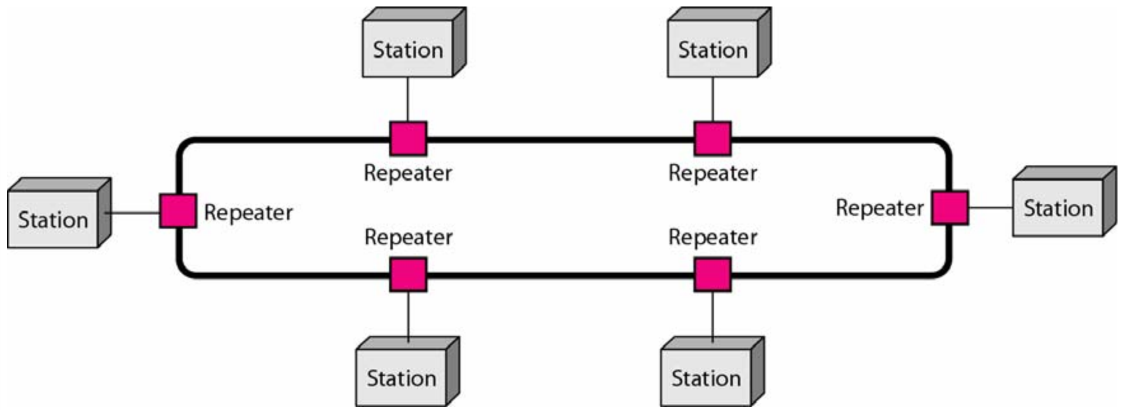


Bus



Star





Ring

Characteristics for LANs with shared medium

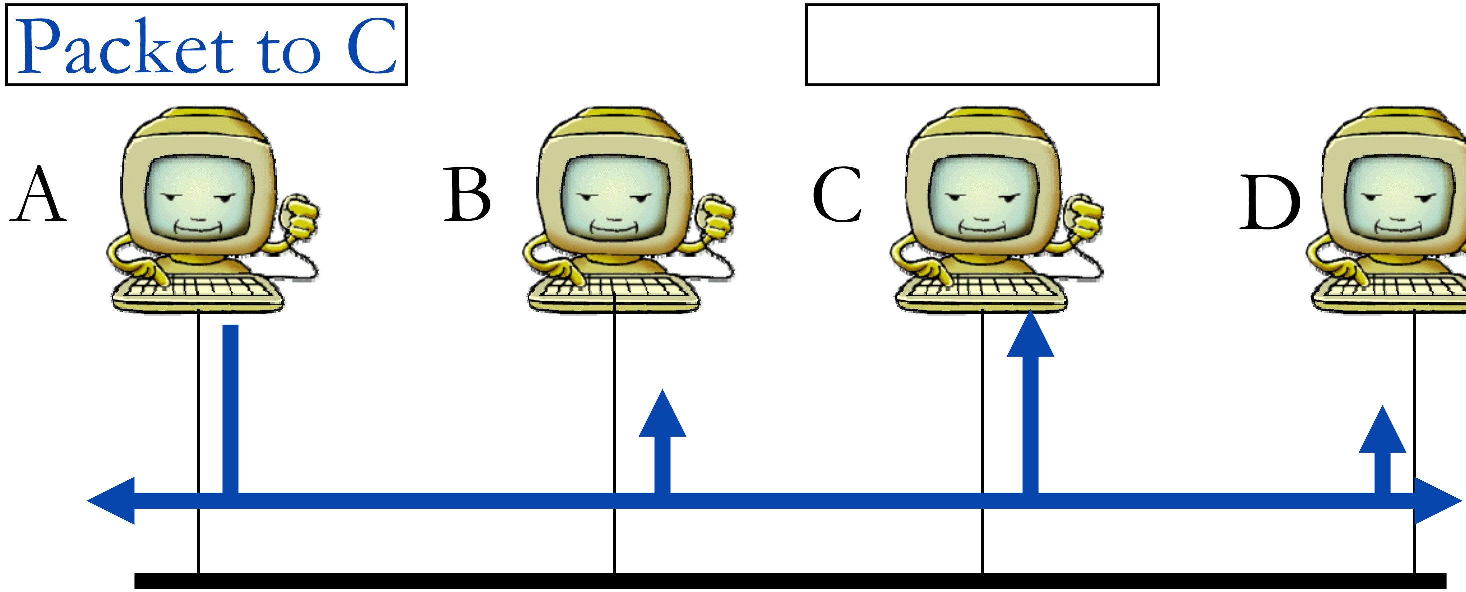
- All data that is transmitted on the link, reaches all stations (broadcast).
- Due to attenuation on the link, the network has a limited size.
- The link can be extended with a *repeater* that amplifies the signal on the link.

LAN addresses

In a network, all stations need an address so that the data can reach the right destination.

All computers connected to standardized LANs have a unique physical address (called MAC-address).

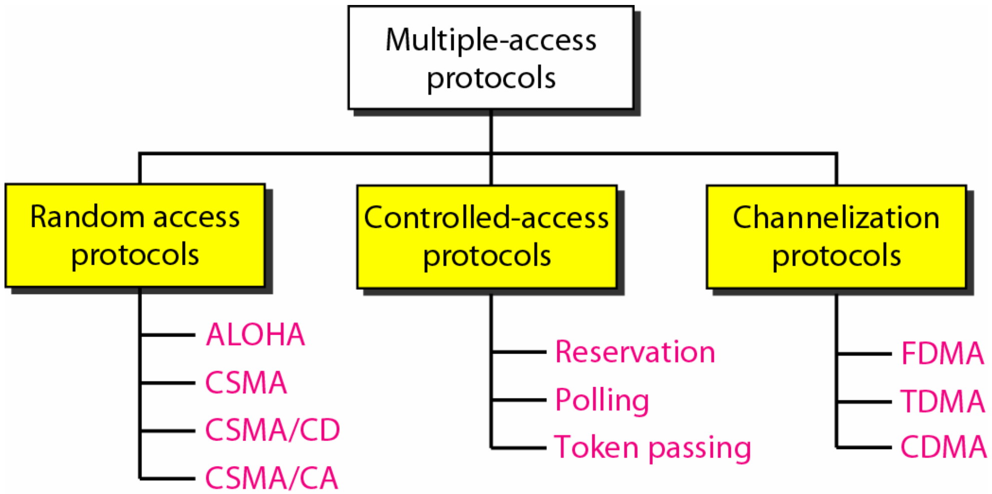
Data transfer on a shared medium



The computer with the right destination address copies the packet and delivers it to the application.

Multiple-Access Protocol

- All computers in a multiple access network, need to have the same rules for sending and receiving data.
- This is called a Multiple-Access Protocol (or medium access control method).

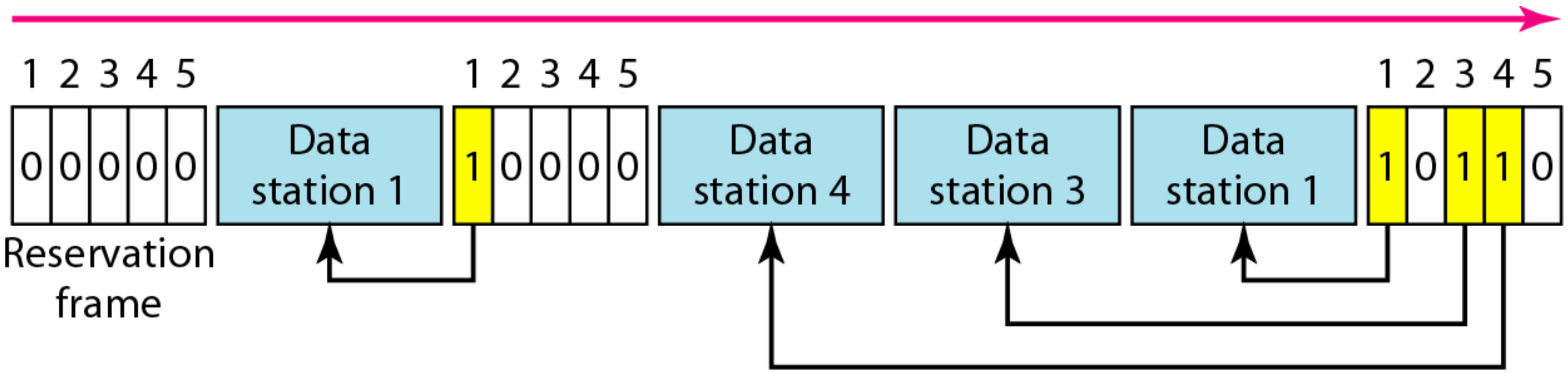


Controlled access methods

In controlled access, the stations consult one another to find which station has the right to send. A station cannot send unless it has been authorized by other stations.

Reservation access method

In the reservation method, time is divided into intervals, and a reservation frame precedes the data frames. A station needs to make a reservation before sending data.



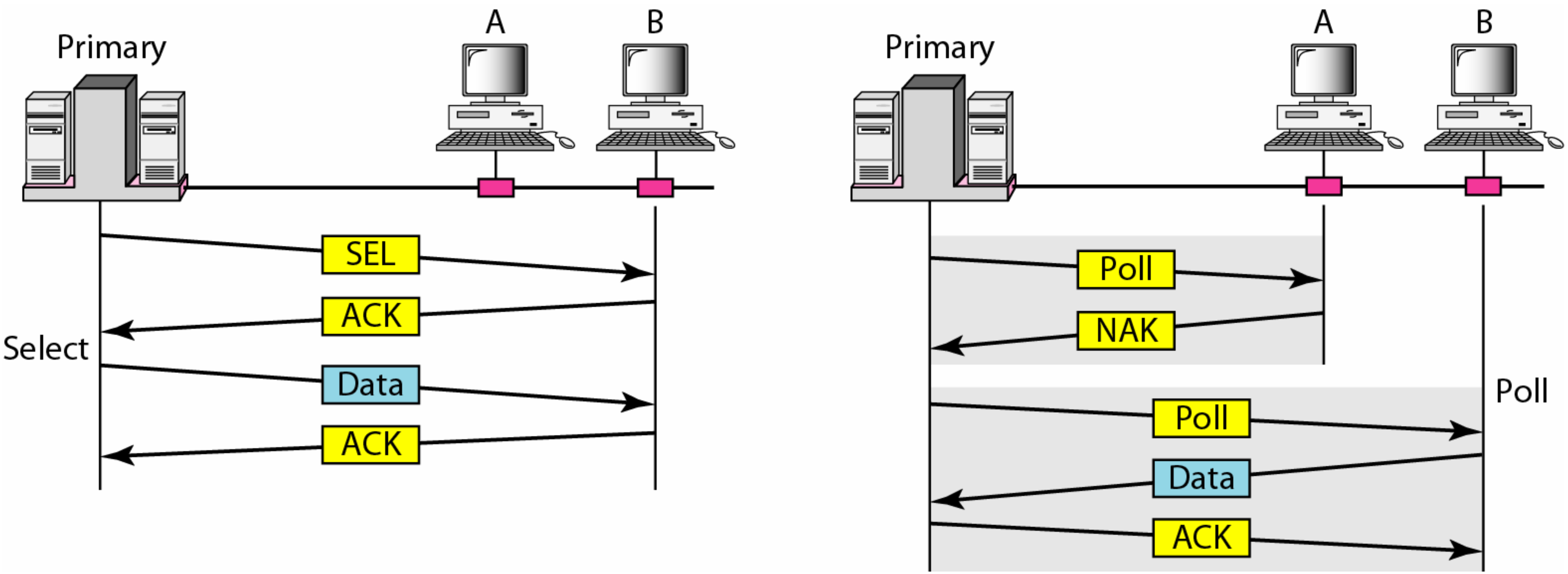
Polling

In Polling, one station is designated as a Primary Station (Master) and the other stations are Secondary Stations (Slaves).

The primary station controls the link, and the secondary stations follow its instructions. All data exchange must be made through the primary station.

If the primary station has anything to send, it uses a Select function. If it wants to receive data it uses a Poll function.

Poll and Select functions



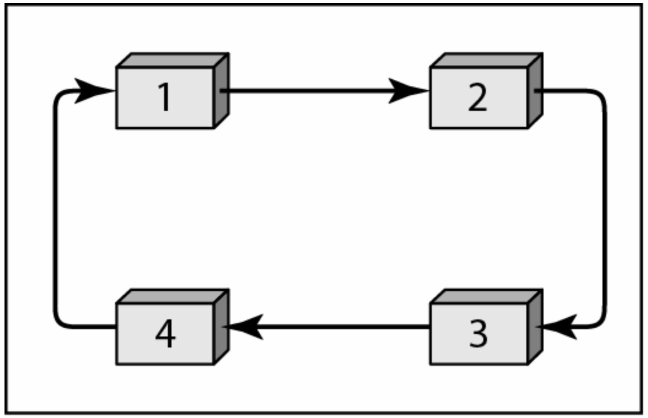
Token Passing

In Token passing the stations are organized in a logical ring. Each station has a predecessor and a successor.

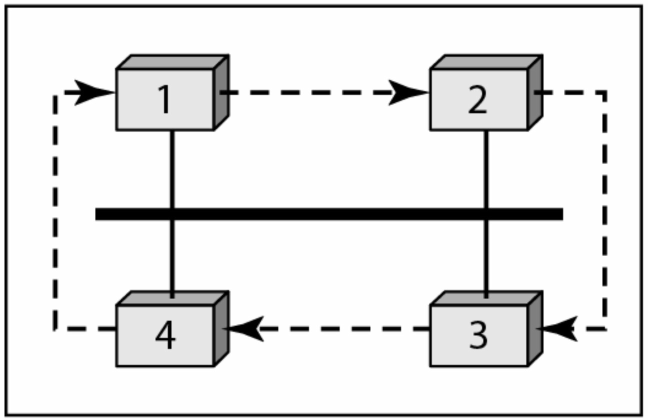
A special packet, called a token, circulates through the ring. The possesion of the token gives a station the right to access the link and send data.

A station can only possess the token for a certain time, then it must release the token and pass it on.

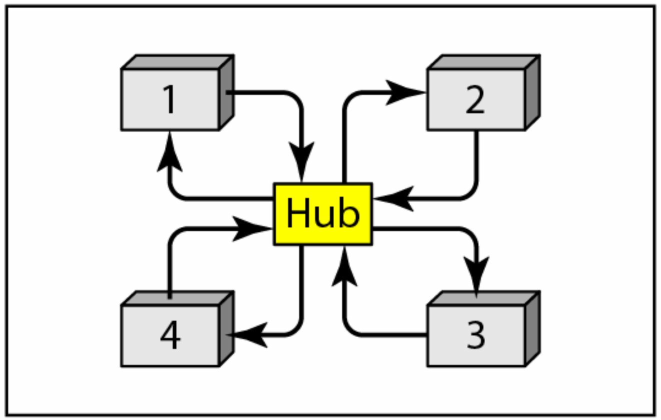
Logical ring and physical topology



a. Physical ring



c. Bus ring



d. Star ring

Random access methods

In random access or contention methods, no station is superior to another station and none is assigned the control over another. No station permits, or does not permit, another station to send.

At each instance, a station that has data to send uses a procedure defined by the protocol to make a decision on whether or not to send.

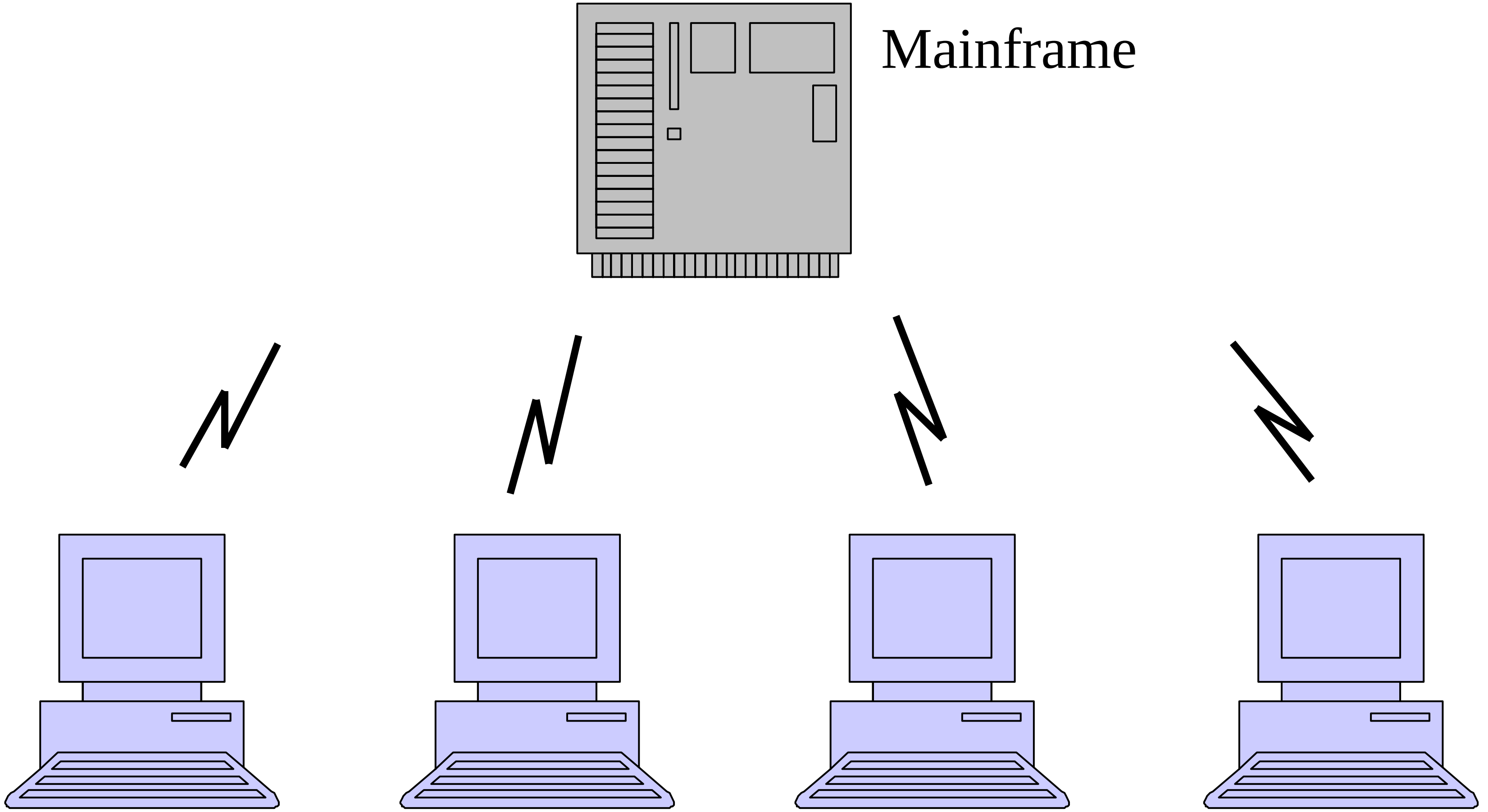
ALOHA

ALOHANET was devloped by the University of Hawaii already in 1970.

ALOHANET was a wireless LAN (one of the first).

The multiple-access method in ALOHANET is called ALOHA.

ALOHANET



Pure ALOHA

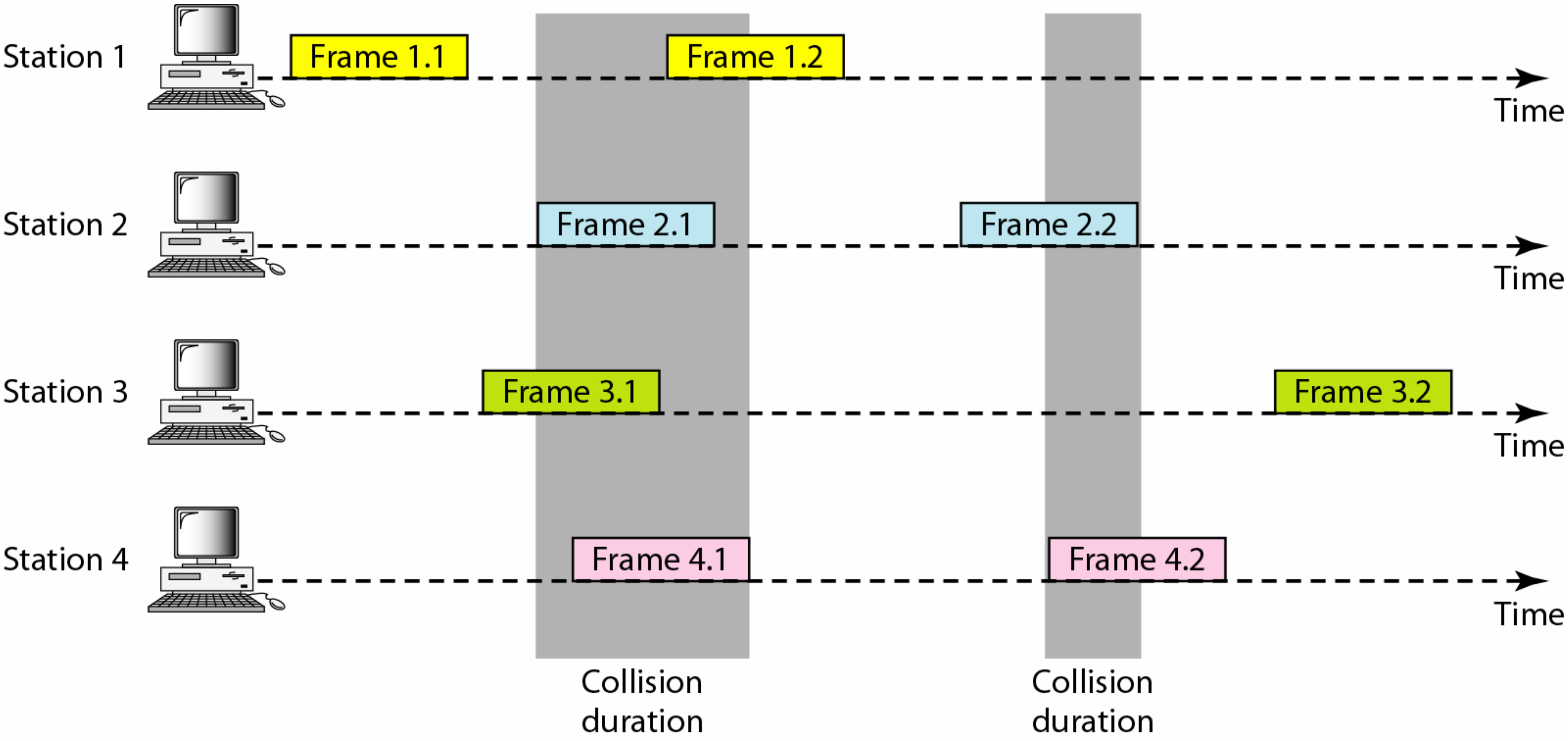
The stations share one frequency band. The mainframe sends data on another frequency (broadcast channel).

A station sends a frame whenever it has a frame to send.

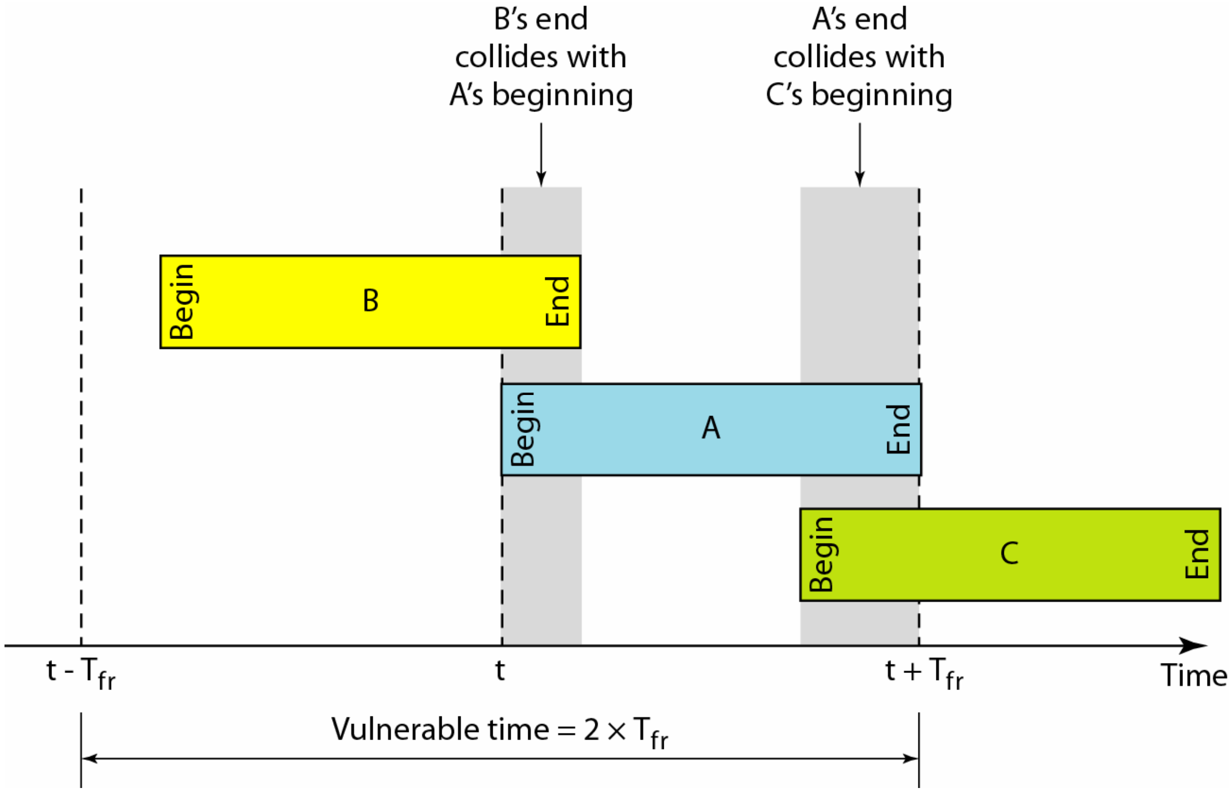
If the station receives an ACK from the mainframe on the broadcast channel, the transmission is successful.

If not, the frame needs to be retransmitted.

Collisions in Pure ALOHA



Performance in Pure ALOHA



The maximal throughput becomes 18% of the link's capacity!

Slotted ALOHA

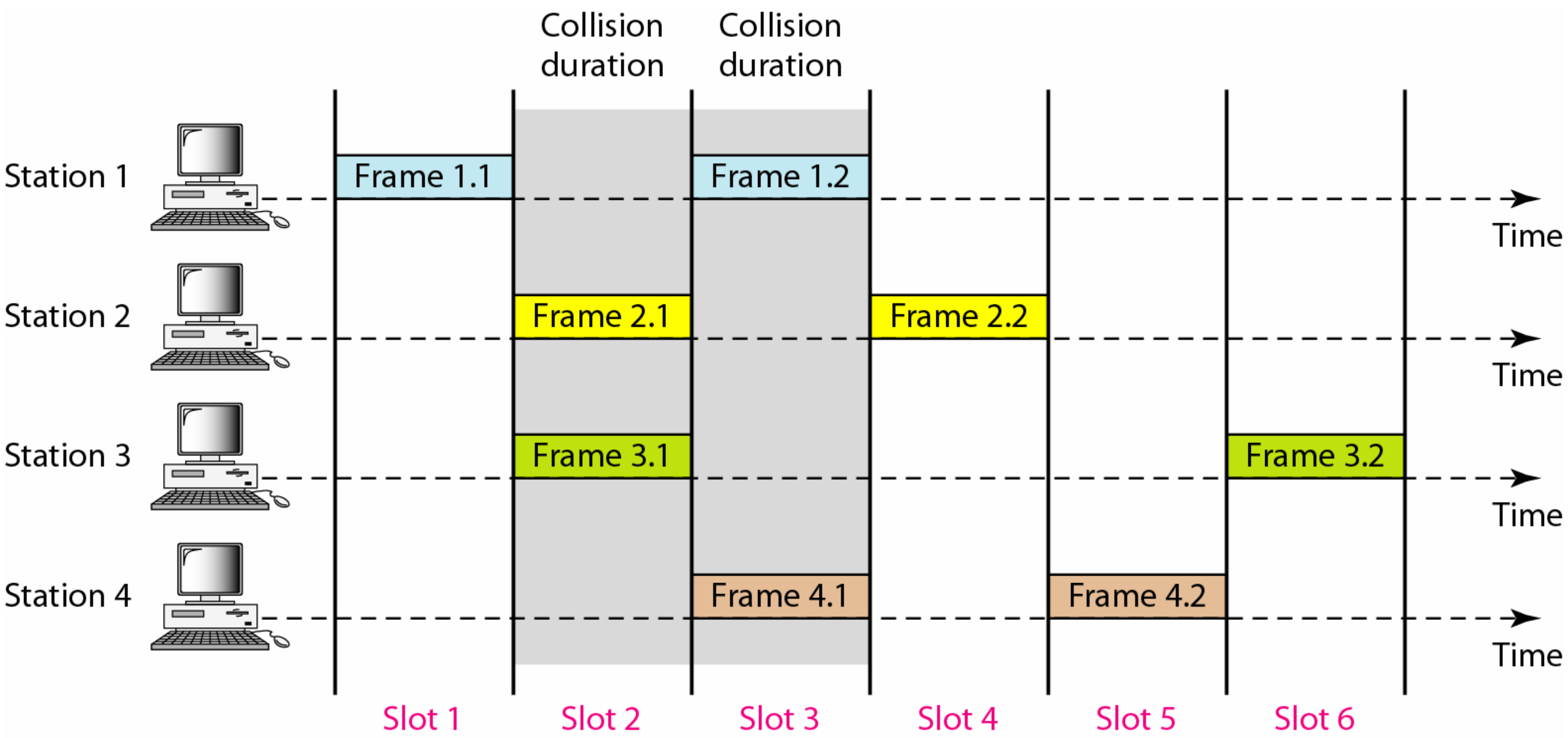
In Slotted ALOHA, the time is divided into slots, where each slot contains one frame.

A station can only send in the beginning of a slot.

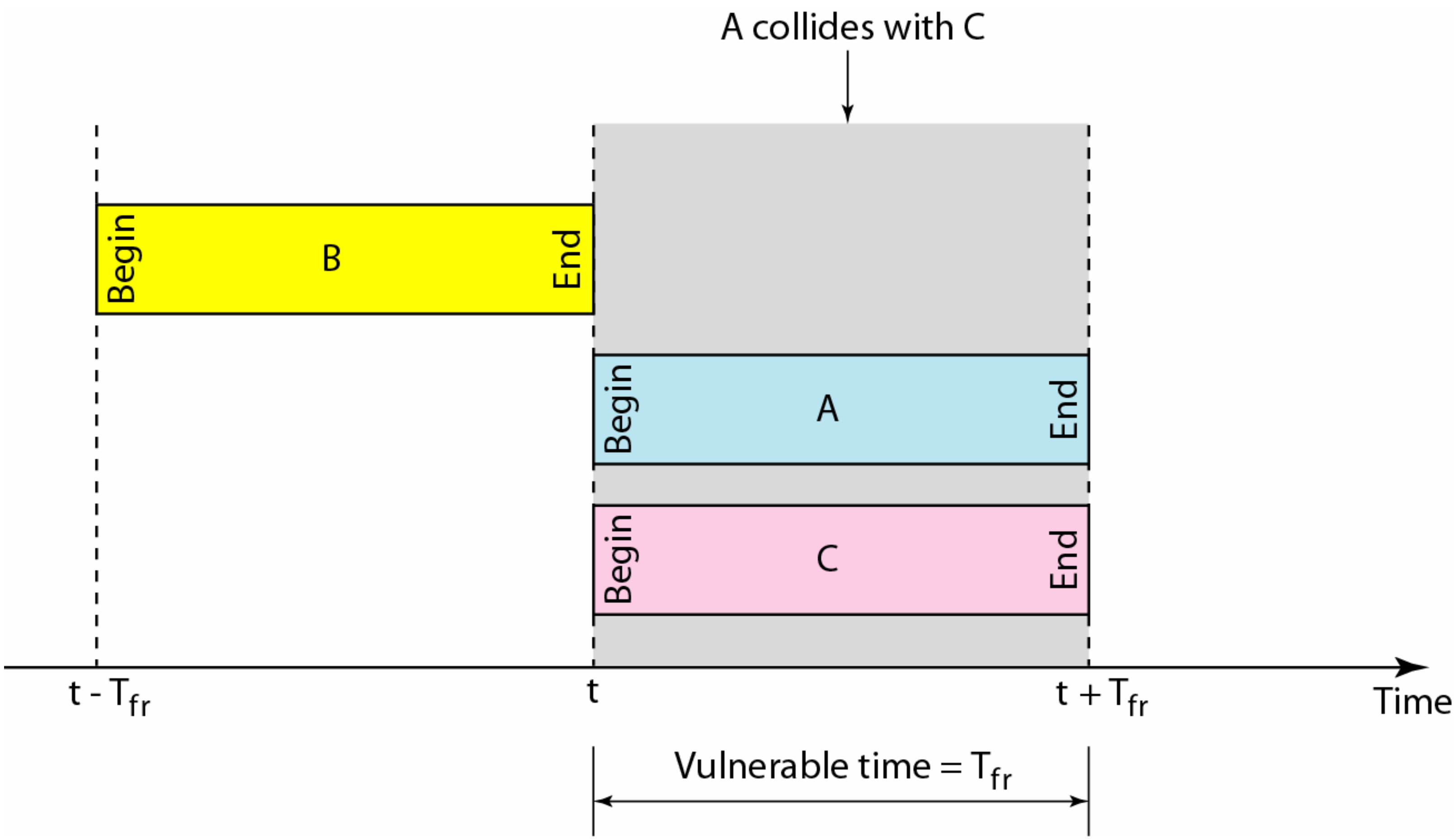
This means that the vulnerable time is reduced to one-half.

The maximum throughput is 36% of the link capacity.

Frames in Slotted ALOHA



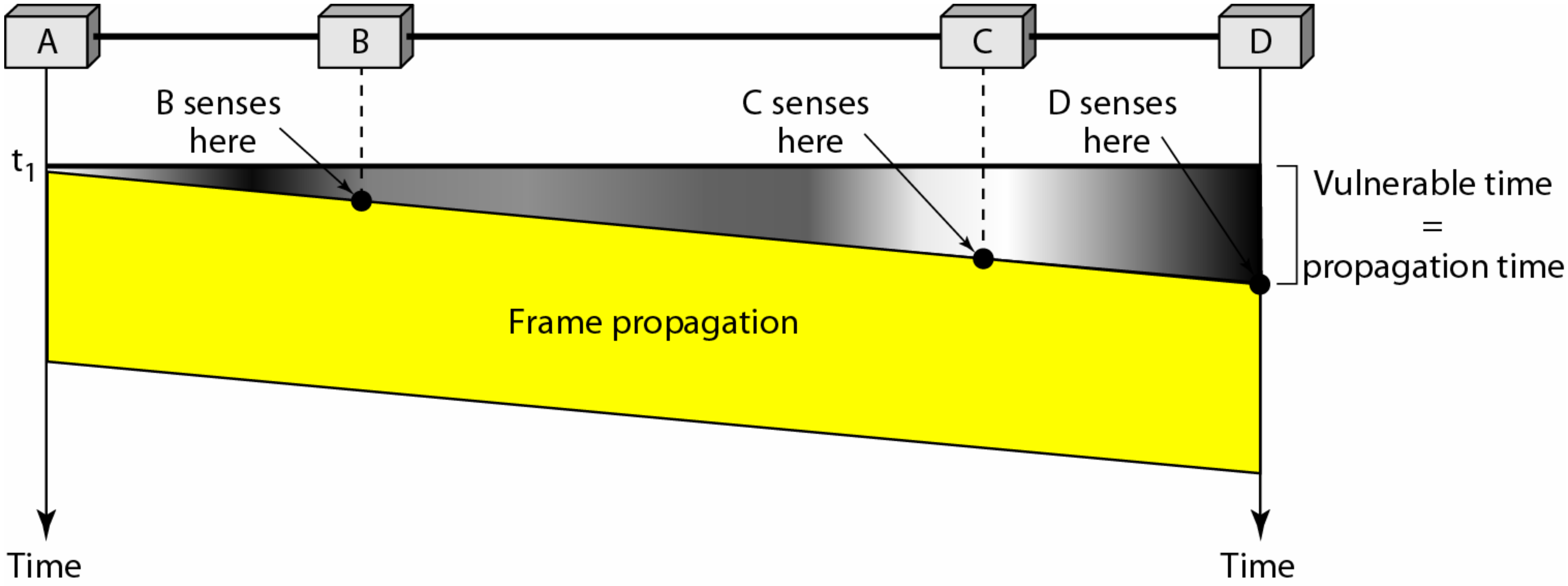
Colllisions in Slotted ALOHA



Carrier Sense Multiple Access (CSMA)

- Before a station sends data, it ”listens” (senses) to the medium.
- If the medium is occupied, the station waits with the transmission.
- The vulnerable time is the propagation time (i.e. the time it takes for the signal to propagate from one of the medium to the other).

Vulnerable time in CSMA

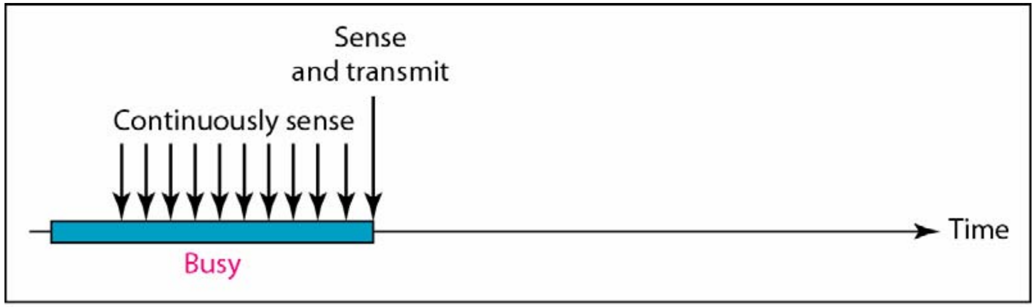


Persistence methods

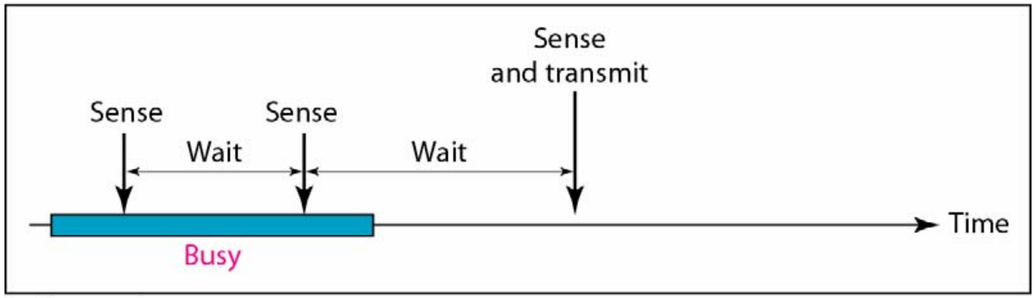
The persistence method decides what a station should do when it senses that the channel is busy.

- 1-persistent: As soon as the channel becomes idle, the station transmits its data.
- Nonpersistent: The station waits a random amount of time and then senses the channel again.
- p-Persistent: The station transmits with probability p. With probability 1-p, it waits for a time period.

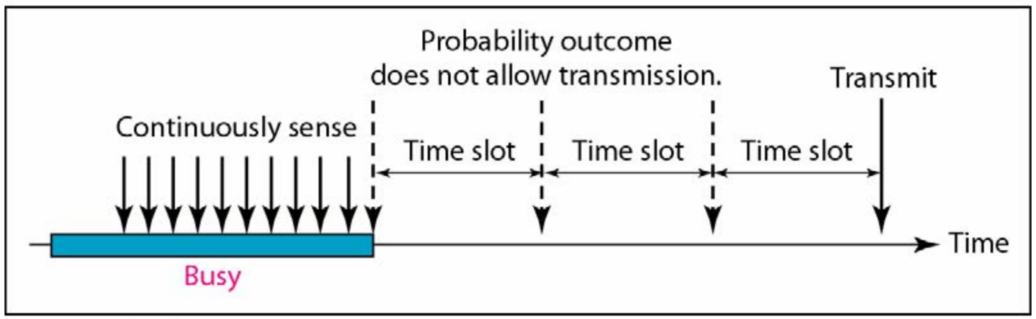
Behavior of the persistence methods



a. 1-persistent



b. Nonpersistent



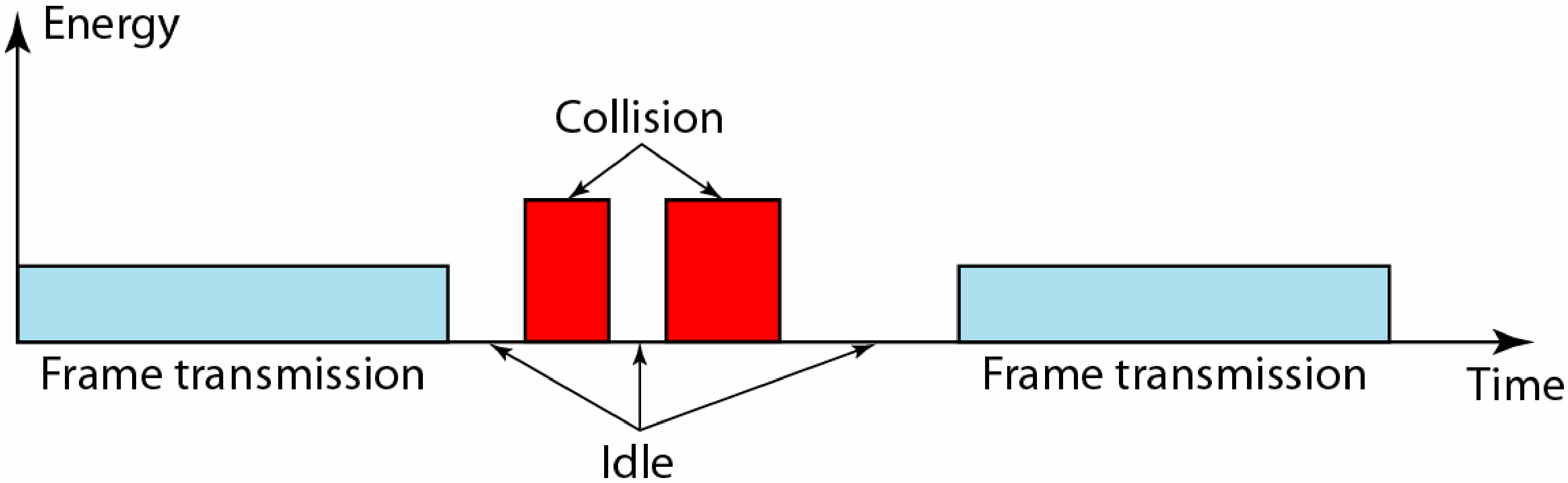
c. p-persistent

CSMA with Collision Detection (CSMA/CD)

The original CSMA method does not specify the procedure following a collision.

- CSMA/CD was developed to better handle collisions.
- After a station monitors the medium after it sends a frame to see if the transmission was successful. A station that detects a collision, immediately aborts the transmission and sends a jamming signal.

Energy levels on link



Minimum frame size

A sending station must be able to detect a collision *before* transmitting the last bit of a frame.

Therefore, the frame transmission time must be *at least* two times the maximum propagation time on the link, so that the colliding signal can propagate back to the sending station before the last bit is transmitted.

CSMA with Collision Avoidance (CSMA/CA)

- CSMA/CD was developed for wired networks that have low attenuation. Therefore, the energy level during a collision can easily be detected.
- In a wireless network much of the energy is lost in transmission. A collision may add only 5-10% additional energy, which is not useful for effective collision detection as in CSMA/CD.
- Therefore, CSMA/CA tries to *avoid collisions*.

Collision avoidance strategies

CSMA/CA has three collision avoidance strategies:

1. Interframe space (IFS)
2. Contention Window
3. Acknowledgment

Interframe space

A station does not send immediately after finding the medium idle.

Instead it waits a period of time, called Interframe space (IFS), since a distant station may have already started transmitting.

If, after the IFS time the channel is still idle, the station can send.

Contention window

The contention window is an amount of time divided into slots.

A station that is ready to send chooses a random number of slots as its wait time.

During the wait time, the station monitors the channel. If the channel is found busy, the timer is stopped and then restarted when the channel is idle.

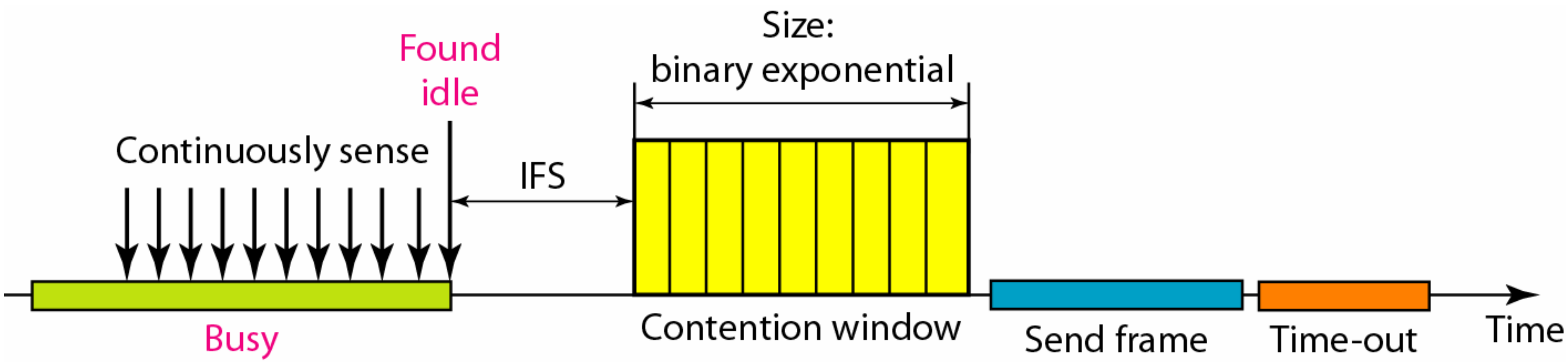
Acknowledgments

When a station has sent a frame, there is a time-out on the channel (no other station can send a frame).

During this time-out the receiving station sends an ACK if the data is received correctly.

If the sending station has not received an ACK within the time-out period, the data is assumed to be lost (either due to collision or bit errors).

Collision avoidance in CSMA/CA



Other multiple-access protocols for wireless networks

In wireless networks, three other multiple-access protocols are also used:

- Frequency-Division Multiple Access (FDMA)
- Time-Division Multiple Access (TDMA)
- Code-Division Multiple Access (CDMA)

These multiple-access protocols works as the multiplexing techniques described before.